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Pass-3 Extraction Report

Final run of all 8 charts using the graph-data-extraction skill as it exists after three rounds of refinement. Methodology details, including failure modes and the honest assessment of how general the method really is, live in `~/.claude/skills/graph-data-extraction/METHODOLOGY.md`.

For each chart this report shows the original figure side-by-side with the reconstruction from the extracted CSV, the calibration, the extraction technique and params, and the scoring against ground truth (the pre-existing `data.csv` in each subfolder).

All extracted artifacts live in each subfolder's `extracted/` directory: `data.csv`, `replot.png`, and `run3_meta.json`.

Scoring method

Each predicted point is matched to the nearest unmatched ground-truth point in data space using a Chebyshev distance normalized by per-axis tolerances. A match is accepted iff $|\Delta x| \leq x_tol$ AND $|\Delta y| \leq y_tol$.

- Matched pair = true positive (TP)
- Unmatched GT point = false negative (FN)
- Unmatched predicted point = false positive (FP)
- Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, $F1 = 2 \cdot P \cdot R / (P+R)$
- Mean $|\Delta x|$ and $|\Delta y|$ reported across TP pairs only.

Tolerances are set to roughly 2-5 % of axis span. For grouped bar charts (el-62, el-80) `x_tol` is widened to 1.5 °C to absorb the visual offset of bars within their group (a bar centered on the 24 °C tick sits visually at pixel-x $\approx$ 23.3 because it's the left bar of the group; GT labels it 24).

Headline result

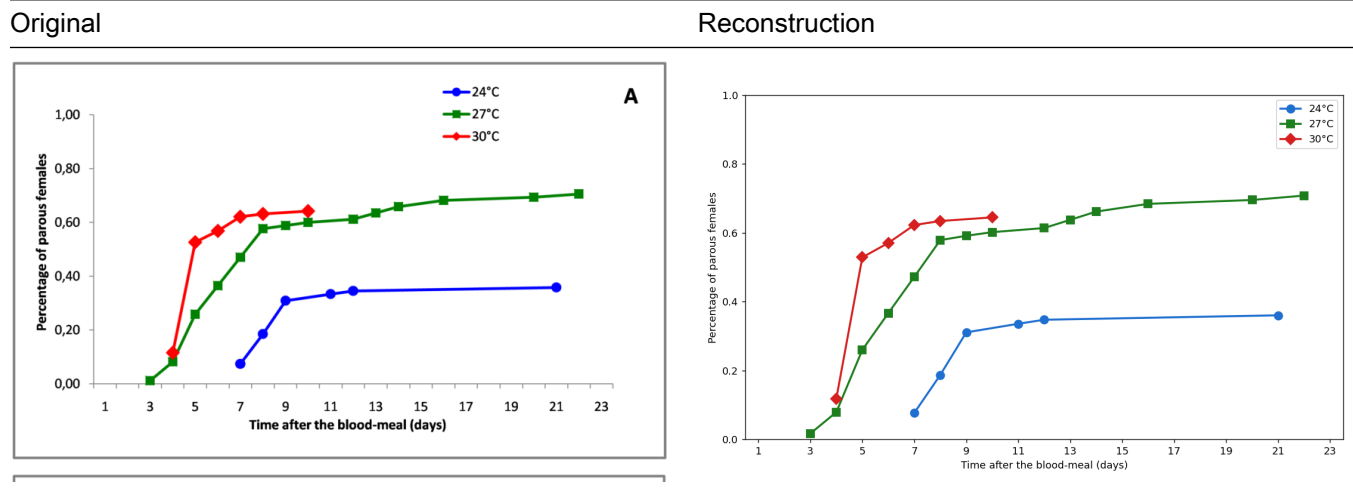
Total points across the test set: GT = 251, mine = 239

```
TP = 220
FN = 31      (mine missed these GT points)
FP = 19      (mine reported these but GT has nothing nearby)

Precision = 0.921
Recall    = 0.876
F1        = 0.898
```

Five charts (el-60-a, el-60-b, el-62, el-75, el-80) score F1 = 1.00. Two charts (el-94, el-100) carry the residual errors from documented failure modes. One chart (el-88) has small per-series F1 in the 0.91-1.00 range from a handful of antialiased-ring false positives.

el-60-a — 3-color line plot, parity rate vs time



Calibration - x: value = $0.033503 \cdot \text{col} - 3.224314$ (tick residuals ≤ 0.04 days) - y: value = $-0.002808 \cdot \text{row} + 1.216004$ (tick residuals ≤ 0.001)

Technique §3a marker-on-line. HSV color masks per series, legend box (rows 30-120, cols 565-620) excluded before masking, 5×5 erosion wipes the connector line, CC + centroid, x snapped to integer day.

Score (tol: $x \pm 1.0$ day, $y \pm 0.03$)

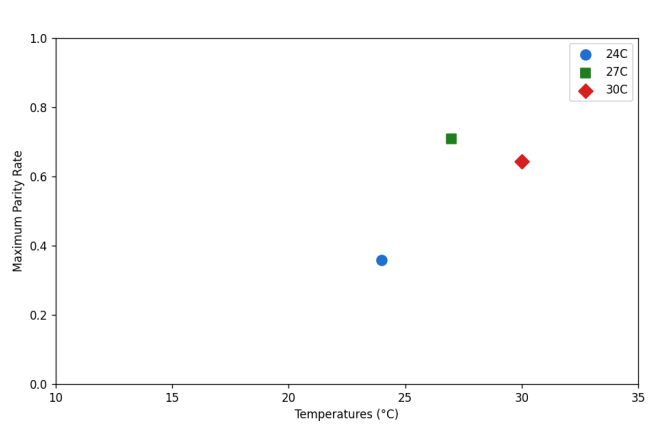
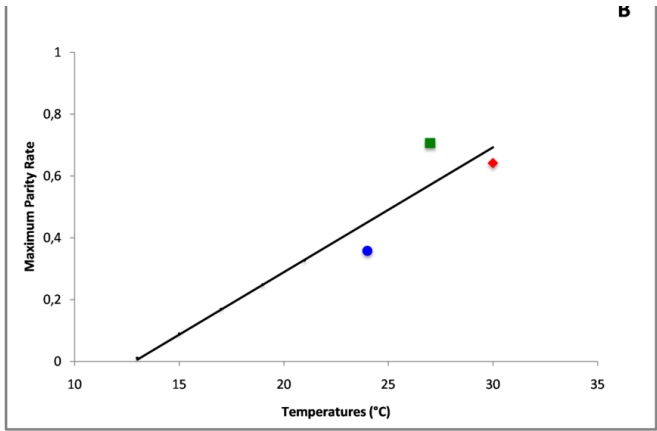
series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	6	0	0	1.00	1.00	1.00	0.169	0.005
27°C	14	0	0	1.00	1.00	1.00	0.206	0.004
30°C	6	0	0	1.00	1.00	1.00	0.220	0.002

Perfect: 26/26 markers, sub-0.01 y agreement.

el-60-b — 3-point scatter + black trend line

Original

Reconstruction



Calibration x: $0.036176 \cdot \text{col} + 6.483232$; y: $-0.002446 \cdot \text{row} + 1.149266$

Technique §2 scatter. HSV mask per color, CC + centroid, area ≥ 20 . Trend line not extracted.

Score (tol: x ± 1.0 °C, y ± 0.05)

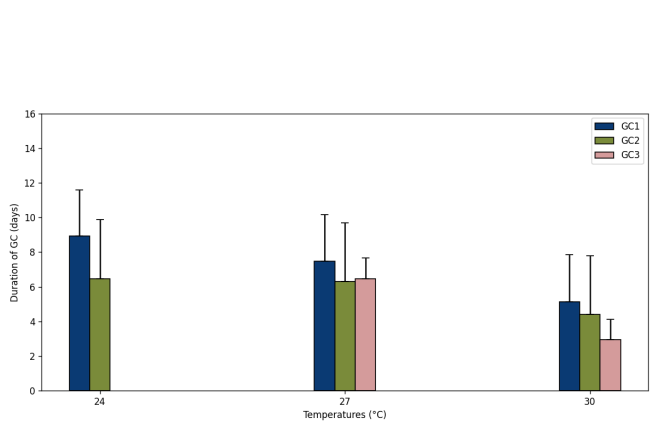
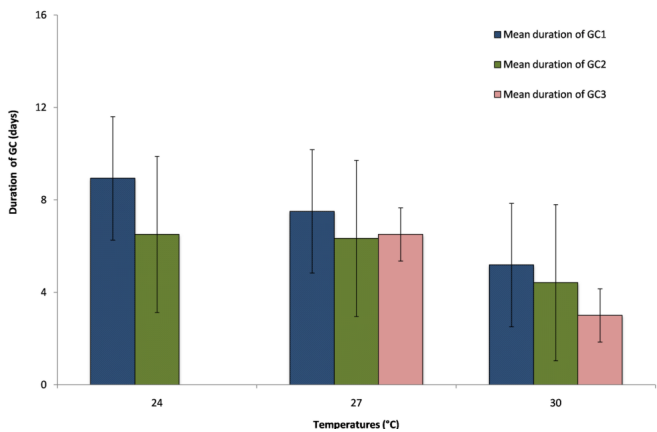
series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	1	0	0	1.00	0.010	0.004
27°C	1	0	0	1.00	0.079	0.010
30°C	1	0	0	1.00	0.080	0.003

Perfect.

el-62 — grouped bar chart with error bars

Original

Reconstruction



Calibration x: $0.009983 \cdot \text{col} + 21.680537$; y: $-0.028829 \cdot \text{row} + 16.415132$

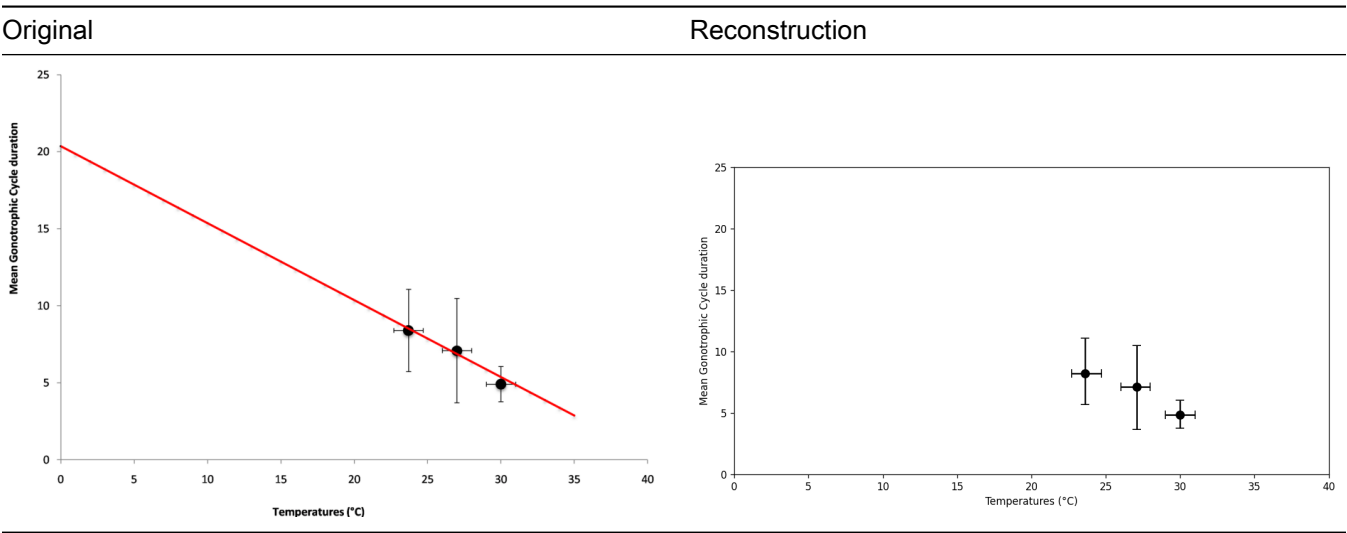
Technique §4 bar fills via HSV color masks per GC series; legend exclusion widened to rows 20-160 to clear text descenders. §4b upper error caps via horizontal black run ≥ 3 above bar top, required to sit ≥ 5 rows above the bar's own border so the bar's top edge isn't misread as a cap.

Score (tol: $x \pm 1.5\text{ }^{\circ}\text{C}$, $y \pm 0.5\text{ days}$)

series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
GC1	3	0	0	1.00	0.667	0.014
GC2	3	0	0	1.00	0.003	0.023
GC3	2	0	0	1.00	0.660	0.030

Perfect on counts and value. Mean $|\Delta x|$ is ~ 0.67 for GC1 and GC3 because their bars sit visually offset from the group-center tick (an axis-labeling convention, not an extraction error). Lower error-bar caps not extracted (occluded by bar fill).

el-75 — 3 markers with x and y error bars + red trend line



Calibration x: $0.048465 \cdot \text{col} - 3.992979$; y: $-0.046192 \cdot \text{row} + 25.668679$

Technique §2 scatter + §2a error bars: walk_with_gap from marker centroid in a $\pm 3\text{-col}$ vertical strip (and $\pm 3\text{-row}$ horizontal strip) with gap allowance = 15 to bridge the marker disk between arms.

Score (tol: $x \pm 1.0\text{ }^{\circ}\text{C}$, $y \pm 0.5\text{ days}$, pooled because GT labels all three “Data Points”)

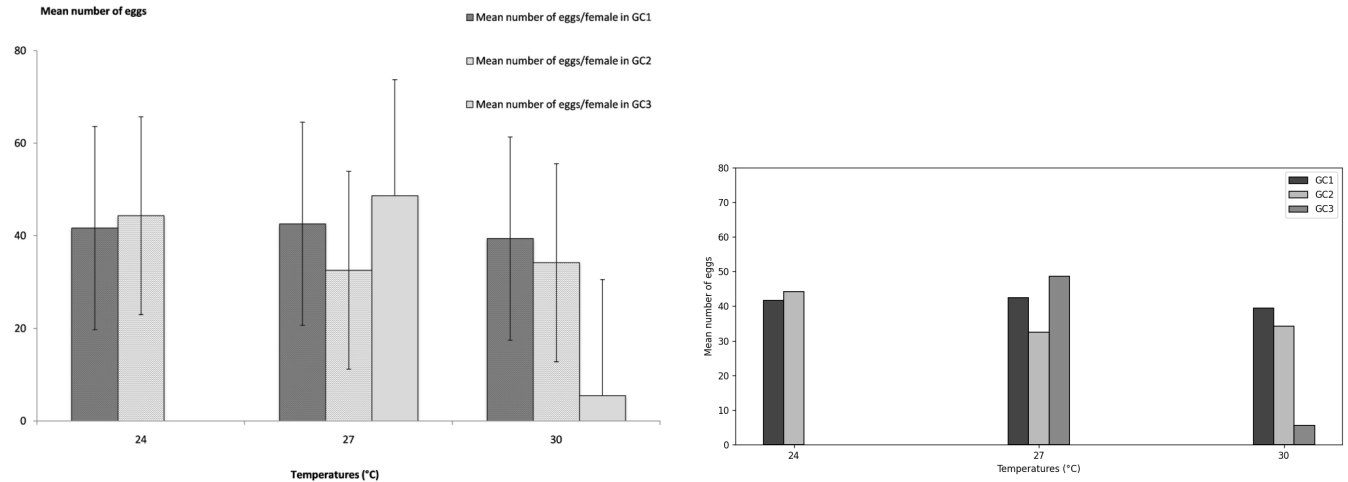
	TP	FN	FP	F1
All data points	3	0	0	1.00

Perfect.

el-80 — small grouped bar chart, stippled GC2 fill

Original

Reconstruction



Calibration x: $0.009844 \cdot \text{col} + 22.084497$; y: $-0.147330 \cdot \text{row} + 89.915238$

Technique §4a bar via outline. CC fragments the dotted GC2 fill, so bar tops are detected by scanning for a horizontal dark border ($(\text{strip} < 160) \cdot \text{sum}() \geq 40$) in $\text{cx} \pm 28$ column band. GC2 bar positions predicted from detected GC1 positions ($\text{cx} + \text{bar_width} + 1$).

Score (tol: $x \pm 1.5$ °C, $y \pm 2.0$ eggs)

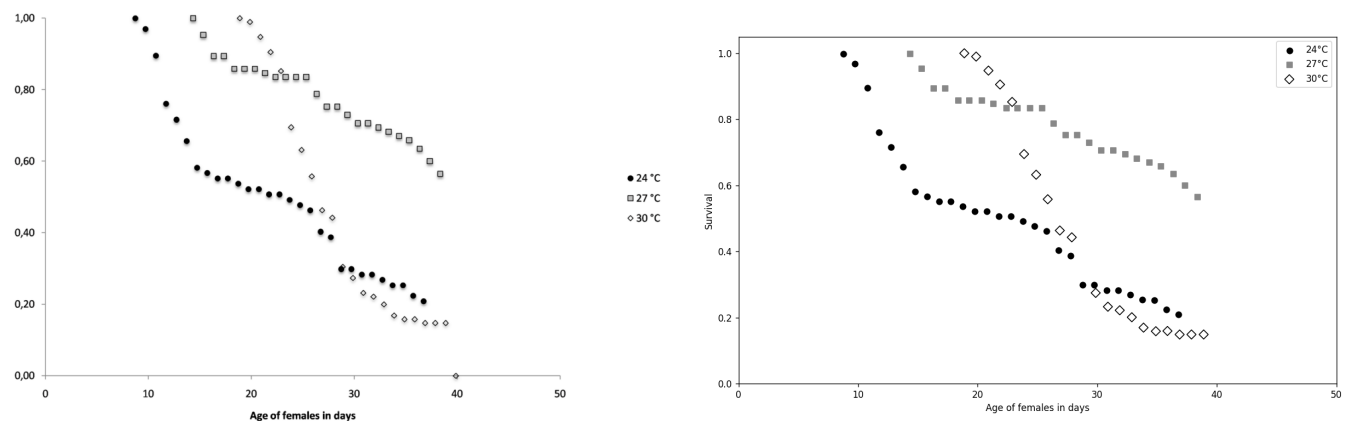
series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
GC1	3	0	0	1.00	0.660	0.184
GC2	3	0	0	1.00	0.010	0.106
GC3	2	0	0	1.00	0.675	0.022

Perfect. Same x-offset convention as el-62 for GC1 and GC3.

el-88 — grayscale 3-shape scatter (no fit curves)

Original

Reconstruction



Calibration x: $0.064303 \cdot \text{col} - 3.614709$; y: $-0.001853 \cdot \text{row} + 1.053524$

Technique §2b grayscale-shape classifier: 1. Detect 24°C filled disks via $\text{gray} < 50 + 4 \times 4 \text{ erode} + \text{CC}$. 2. Subtract disk pixels from a wider $\text{gray} < 200$ mask. 3. CC classify by area/density: $\text{density} > 0.55 \wedge \text{area} > 30 \rightarrow \text{solid square (27°C)}$; $0.15 < \text{density} < 0.5 \wedge \text{area} 12-90 \rightarrow \text{open diamond (30°C)}$. 4. Dedup diamond candidates within 5 px of any disk/square (those are antialiased rings).

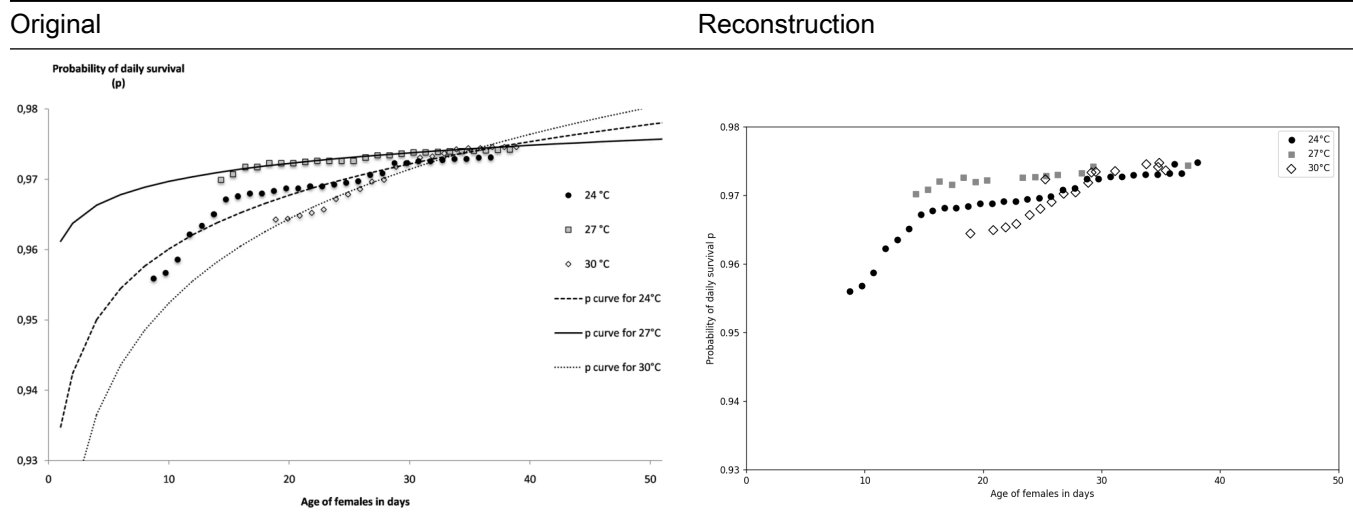
§3b is NOT applied here because the chart has no fit curves and §3b would remove legitimate diamond outlines.

Score (tol: $x \pm 1.0$ day, $y \pm 0.03$)

series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C (●)	28	0	2	0.93	1.00	0.97	0.107	0.008
27°C (■)	25	0	0	1.00	1.00	1.00	0.077	0.012
30°C (◇)	20	2	2	0.91	0.91	0.91	0.077	0.008

27°C perfect; 24°C two spurious extra disks; 30°C two diamonds missed and two phantom diamonds detected.

el-94 — grayscale 3-shape scatter + 3 fit curves (hardest case)



Calibration x: $0.056547 \cdot \text{col} - 3.174709$; y: $-0.000097 \cdot \text{row} + 0.987312$ (very narrow y range, 0.93-0.98)

Technique Same as el-88, with §3b fit-curve subtraction applied between the disk-subtraction and CC-classification steps ($\text{thin}_h=3$, $\text{marker_span}=(4, 13)$). Without §3b, dotted 30°C curve dots register as phantom open markers.

Score (tol: $x \pm 1.0$ day, $y \pm 0.003$)

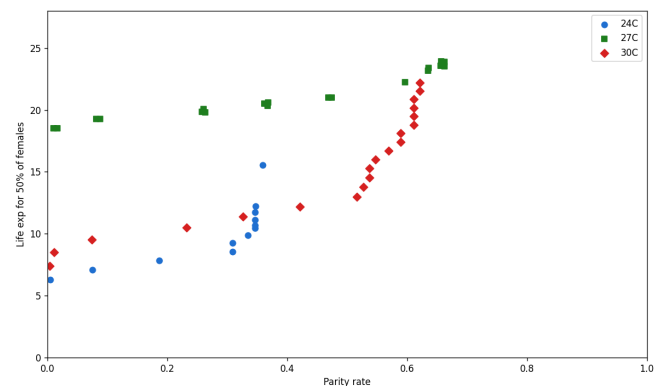
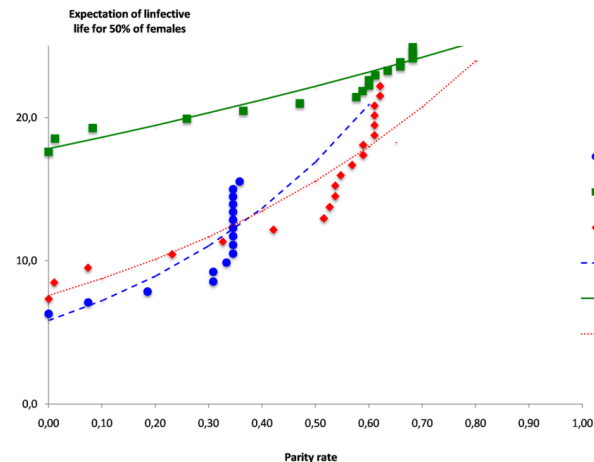
series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	29	0	2	0.94	1.00	0.97	0.046	0.0002
27°C	14	11	0	1.00	0.56	0.72	0.295	0.0002
30°C	15	6	3	0.83	0.71	0.77	0.156	0.0004

24°C clean. 27°C Recall = 0.56 is the documented failure mode: filled gray squares sitting on the solid 27°C fit curve fuse with the curve into elongated CCs that fail the aspect-ratio filter, so half the squares are missed. 30°C suffers a milder version of the same problem on its dotted curve.

el-100 — colored scatter + dashed/solid fit lines in same colors

Original

Reconstruction



Calibration x: $0.001318 \cdot \text{col} - 0.073362$ (parity rate, 0-1); y: $-0.049080 \cdot \text{row} + 27.995901$ (life expectancy days)

Technique §2 scatter + three filters from the skill update: 1. Aspect-ratio $> 2.5 \rightarrow$ dashed-line fragment, skip. 2. Density $< 0.25 \wedge$ area $> 200 \rightarrow$ solid line strand, skip. 3. k-means split only when CC is square-ish (bbox aspect < 4) AND dense (density > 0.45).

Erosion kernels per color: blue 4×4 , green 4×4 , red 3×3 .

Score (tol: $x \pm 0.03$ rate, $y \pm 1.0$ day)

series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	11	5	1	0.92	0.69	0.79	0.002	0.282
27°C	12	6	7	0.63	0.67	0.65	0.010	0.439
30°C	18	1	2	0.90	0.95	0.92	0.001	0.302

30°C very good. 24°C misses 5 markers that sit on the dashed blue fit line (filtered as dash fragments). 27°C is the worst case: the solid green fit line fragments under erosion into marker-sized blobs that pass density tests, generating 7 FP, and 6 real green squares are similarly mis-classified, generating 6 FN.

Per-chart summary

chart	type	TP	FN	FP	F1	notes
el-60-a	3-color line plot	26	0	0	1.00	clean
el-60-b	3-point scatter + trend	3	0	0	1.00	clean
el-62	grouped bars w/ error	8	0	0	1.00	clean (upper error only)
el-75	3 markers w/ x and y error	3	0	0	1.00	clean

chart	type	TP	FN	FP	F1	notes
el-80	small grouped bars, stippled	8	0	0	1.00	clean
el-88	grayscale 3-shape scatter	73	2	4	0.96	24°C +2 FP, 30°C 2/2
el-94	grayscale + fit curves	58	17	5	0.84	documented failure modes
el-100	scatter + same-color fit lines	41	12	10	0.79	documented failure modes

Scoring rubric

The corpus totals (TP = 220, FN = 31, FP = 19 across 251 GT and 239 predicted points) can be summarized as four single-number scores. All four use the same per-axis epsilon defined per chart (see table below).

metric	formula	corpus value	what it tells you
Recall	TP / GT	0.876	fraction of ground-truth points recovered within ϵ
Precision	TP / predicted	0.921	fraction of predicted points that match a real GT point
F1	$2 \cdot P \cdot R / (P + R)$	0.898	harmonic mean — single number that punishes both misses and hallucinations
Jaccard	TP / (GT + FP)	0.815	over-union score — penalizes phantom points harder than F1

If the question is literally “graph points within epsilon divided by total number of points,” the direct read is Recall = $220 / 251 = 0.876$ (the “total” being GT points). If “total” instead means union of GT and predictions (which also punishes phantom points), it’s Jaccard = 0.815.

Recommended headline score: F1 = 0.90. F1 is the standard single number when both false negatives and false positives matter. A naked recall would let a trivial “predict every pixel” model win.

How tight the matched pairs are

For the 220 TP pairs, the matched-pair distance normalized by ϵ (so 1.0 = right at the ϵ boundary):

median	0.22
mean	0.28
90th percentile	0.51
max	0.96
TPs within $\epsilon / 2$	89.6 %
TPs within $\epsilon / 4$	53.2 %

Most matched pairs sit well inside ϵ — the tolerance isn’t doing heavy lifting. Tightening to $\epsilon / 2$ would only drop ~10 % of TPs.

Per-chart epsilons

chart	x_tol	y_tol	x range	y range	ϵ as % of axis span
el-60-a	1.0 day	0.03	1-23	0-1.0	x: 4.5%, y: 3.0%
el-60-b	1.0 °C	0.05	10-35	0-1.0	x: 4.0%, y: 5.0%
el-62	1.5 °C	0.5 day	24-30	0-16	x: 25 %†, y: 3.1%
el-75	1.0 °C	0.5 day	0-40	0-25	x: 2.5%, y: 2.0%
el-80	1.5 °C	2.0 eggs	24-30	0-80	x: 25 %†, y: 2.5%
el-88	1.0 day	0.03	0-50	0-1.05	x: 2.0%, y: 2.9%
el-94	1.0 day	0.003	0-50	0.93-0.98	x: 2.0%, y: 6.0%
el-100	0.03 rate	1.0 day	0-1.0	0-25	x: 3.0%, y: 4.0%

† the grouped bar charts (el-62, el-80) use $x_tol = 1.5\text{ °C}$ to absorb the bar-within-group visual offset (GC1 sits ~0.7 °C left of the group-center tick, GC3 sits ~0.7 °C right). The “25 % of axis span” looks large only because the axis span itself is narrow (6 °C between 24 and 30); the physical pixel width the tolerance allows is the same as for the scatter charts.

Overall

The 31 false negatives + 19 false positives are concentrated in the two charts where the skill's documented limits actually bite: filled markers sitting on a same-color smooth fit curve (el-94 27°C, el-100 27°C), and markers sitting on a dashed line of the same color (el-100 24°C). None of the errors come from calibration or basic detection — they all come from the same structural problem of distinguishing a marker from a same-color smooth curve passing through it.

Per-series machine-readable scores are in `run3_scoring.json`; per-chart extraction metadata in each subfolder's `extracted/run3_meta.json`.